

Documentation for the McDIPPER package

Oscar Garcia-Montero¹

¹*Fakultät für Physik, Universität Bielefeld, D-33615 Bielefeld, Germany*

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I. INTRODUCTION

This is the documentation for the McDIPPER package. It includes a list of the options which can be included in the input file for the creation of initial conditions. For details of the analytical formulas, and the physical meaning of the parameters, please consult the main reference of this work, ref. [1].

II. INPUT OPTIONS

The McDIPPER code uses `.yaml` input files. These files have strict spacing rules for the header and subheader options, each subheader needs to be included the corresponding (sub-)header with exactly *four spaces*. So, each subsequent indentation uses an extra indentation of that size.

Two independent headers are to be included in the first part of the input file. These are the level of terminal output (verbosity) and, anticipating the future improvements, the version of the code (for now only 1.0). These are to be included

- **Version:** Code Version.
- **Logging:** Terminal output header,
 - Followed by the subheader **Verbose**:
 - * **Verbose=0**: No output. Only errors are printed.
 - * **Verbose=1**: Minimal output (suggested).
 - * **Verbose=2**: Minimal event information.
 - * **Verbose=3**: Full EbE output.

The rest of the main headers are

- **General**: General aspects of the collision.
- **Grid**: Parameters for the spatial grid.
- **Model Parameters**: Parameters of the saturation model.
- **Output**: Output commands.

These headers will be detailed in the next sections.

A. General

The general subheader corresponds to the general information needed for the creation of the initial state of the collision. The subheaders available to the user are the following:

- **SqrtsNN**: Collisional energy of the heavy ion collision per nucleon, $\sqrt{s_{\text{NN}}}$, in GeV. Under the energy of the collision, one need to specify the incoming nuclei, which can be done using the following structure:

```
Nucleus1:
  A:  208
  Z:  82
  mode: 0
Nucleus2:
  A:  208
  Z:  82
  mode: 0
```

where **Z**, **A** correspond to the atomic number and mass number relatively. The keyword **mode** corresponds to

- Spherical Nuclei (**mode**: 0) Included are $^{16}_8\text{O}$, $^{40}_{18}\text{Ar}$, $^{63}_{29}\text{Cu}$, $^{129}_{54}\text{Xe}$, $^{197}_{79}\text{Au}$ and $^{208}_{82}\text{Pb}$
- Deformed Nuclei (**mode**: 1) Included are $^{27}_{13}\text{Al}$, $^{63}_{29}\text{Cu}$, $^{129}_{54}\text{Xe}$, $^{197}_{79}\text{Au}$ and $^{238}_{92}\text{U}$
- List Mode (**mode**: 3) the MCDIPPER takes in a list of N_{conf}

These parameters can be found in file **nuclear_data.h**, and are taken from Ref.[2]. Additionally, configurations for deuteron, ^2_1H , and ^3_2He are included in the **nuclear** folder, and can be used in the input file normally by listing **A** and **Z**. More can be included by petition.

If **mode** is set to list mode, extra parameters need to be input.

- **InputFile**: The path to the configurations
- **IsospinSpecified**: if set to **True**, nucleons are flagged as neutrons(0) or protons(1).
- **Configurations**: Number of different nuclear configurations in the file. The format of these configurations lists are different depending on the value of **IsIsospinSpecified**. For **IsIsospinSpecified=False**, the MCDIPPER requires a file with the following format:

$$1. \text{ID}_{conf} \quad 2. x_{nucleon} \quad 3. y_{nucleon} \quad 4. z_{nucleon} \quad (1)$$

If the isospin of the nuclei is specified, the only difference is an extra column with the appropriate value for the respective nucleon:

$$1. \text{ID}_{conf} \quad 2. x_{nucleon} \quad 3. y_{nucleon} \quad 4. z_{nucleon} \quad 5. I_{3,nucleon} + 1/2 \quad (2)$$

where in standard notation $I_3 = \pm 1/2$ for nucleons.

- **Events**: Number of events to be created.
- **GlauberAcceptance**: Nucleons are sampled from the Woods-Saxon distribution using a rejection sampling (see eq. (17) in the main paper). This flag chooses the geometry of the nucleons for that sampling. The default is **Gaussian**.
- **Model**: Saturation model to include in the computation of the energy and charge deposition.
 - GBW model (**Model**: 0)
 - IP-Sat model (**Model**: 1)
- **Impact**: Impact parameter options. There are two modes: fixed value and sampling.
 - **Value**: Fixed value of the impact parameter, has to be given in units of fm, e.g. **Value**: 3.2.
 - For the range sampling, an example is given below,

```
Impact:
  Range: [0, 20]
  Sampling: Quadratic
```

The flag **Sampling** stands for the two possibility for sampling of the absolute value of the impact parameter, **Quadratic** and **Uniform**. The first stands for the *dbb* sampling, while the second stands for simple *db* sampling.

- **PDFs**: User parameters to use with the LHAPDF library.
 - **PDFSet**: This is the PDF parametrization used in the computation of quark quantities, they need to be previously downloaded using the instructions in the LHAPDF webpage. In the main reference we have mostly used the CT18NNLO PDF set.
 - **ForcePositive**: Enforce positivity on the PDF set (**ForcePositive**: 1). This is particularly important for our purposes as the PDF interpolation gets unstable at low virtuality. These low Q^2 pieces don't contribute to the energy and number density, but can increase the relative error of the integration routines.
- **K-Factor**: The free parameter of the MCDIPPER framework. Needs to be fitted (see main paper). For most purposes its inclusion is linear in the gluon observables *except* the multiplicity estimator which is given by a power law of the sum of the energies, $\int d^2\mathbf{x} [\tau K_g \epsilon_g + \epsilon_q]^{2/3}$.

- **Fluctuations:** Enables fluctuations in addition to the default nucleon EbE position (geometrical) fluctuations. Main reference [3?].

- **Thickness_fluct:** If set to **True**, the thickness function's normalization is rescaled by weight p_i , sampled from a distribution with average $\langle p_i \rangle = 1$.
- **Fluct_mode:** Distribution to sample the weights p_i from. Two modes are currently implemented:
 - * **Gamma distribution:**

$$p(p_i; \sigma, \beta) = \frac{1}{\Gamma(\sigma)\beta^\sigma} p_i^{\sigma-1} e^{-p_i/\beta} \quad (3)$$

where the shape parameter σ and scale parameter β are chosen such that $\langle p_i \rangle = 1 \rightarrow \beta = \sigma^{-1}$.

- * **Log-normal distribution:**

$$P(\ln p_i) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{\ln^2 p_i}{2\sigma^2}\right]. \quad (4)$$

Subsequently, the sampled p_i are scaled by the expectation value of the distribution $E[p_i] = e^{\sigma^2/2}$ to keep the average density unchanged.

- **Sigma:** The parameter σ controls the magnitude of thickness fluctuations.
- **Hotspots_fluct:** If set to **True**, the nucleon is split into N_q smaller density regions,

$$t_{p/n}(\mathbf{b}_\perp) = \frac{1}{N_q} \sum_{i=1}^{N_q} p_i t_q(\mathbf{b}_\perp - \mathbf{b}_{\perp,i}), \quad (5)$$

and assume each hotspot presents a Gaussian profile,

$$t_q(\mathbf{b}_\perp) = \frac{1}{2\pi B_q} e^{-\mathbf{b}_\perp^2/(2B_q)}. \quad (6)$$

The hotspot positions $\mathbf{b}_{\perp,i}$ are sampled from a two-dimensional Gaussian distribution whose width is denoted by B_{qc} . We set $B_{qc} = \frac{B_G - B_q}{1 - 1/N_q}$ to reproduce the nucleon width B_G .

- **Bq:** Size of the hotspot density profile.
- **Nq:** The number of hotspots to be sampled per proton/neutron. Default is set to $N_q = 3$.

B. Grid

Under the **Grid** header, all parameters defining the spatial output of the initial state must be given. These are the following.

- **NX:** Number of points along the x direction
- **NY:** Number of points along the y direction
- **NETA:** Number of points along the η_s direction.
- **X_RANGE:** Domain of the x dimension, needs to be given as, for example, $[-12, 12]$
- **Y_RANGE:** Domain of the y dimension, needs to be given as, for example, $[-12, 12]$
- **ETA_RANGE:** Domain of the $x\eta_s$ dimension, needs to be given as, for example, $[-8, 8]$
- **BG:** Effective size of the nucleon, as defined from its thickness function, see eq. (9) from the main paper. Given in units of (fm^2).

C. Model Parameters

- GBW model (**Model**: 0): Controls the parameters of the GBW model, see eq. (14) of the main reference, $Q^2(x) = Q_0^2(x/x_0)^\lambda(1-x)$.
 - **Q02**: Saturation scale, given in units of GeV^2
 - **x0**: Fixing value if value of the saturation scale is known from a model at a specific x , say $Q_0^2 = Q^2(x_0)$. This should be normally specified at small x . In the case of the main reference, the saturation scale is known from a fit, so **x0** is set to 1.
 - **lambda**: Power law for the saturation scale.
 - **XCut**: Large- x cutoff for the GBW model. At $x > \text{XCut}$ the dipole is set to 0. This is included to test the gluon production influence on the observables at large rapidities. Set it to **XCut**= 1 as a default.
- IP-Sat model (**Model**: 1)
 - **Set**: Chooses the parameter set of the IP-Sat fits from ref. [4]. **Set**:1 corresponds to the fits for the charm mass, $m_c = 1.27$, while **Set**:2 corresponds to the fits for $m_c = 4$.
 - **XScaling**: Sets the switch to a geometrical scaling behaviour for the IP-Sat dipole for $x > \text{XScaling}$. We recommend using values between 0.01 – 0.1, as the data becomes sensitive to numerical resolution above those values, and below, the IP-Sat dynamics is washed out.

D. Output

- **path_to_output**: Path to output folder. Default: **out/**
- **run_name**: Output folder name. Default is set to **Run*i***. The code will find the last run (say **Run10**) and set it to the next integer **Run11**.
- **Format**: Output format. Included are three available options:
 - * **Charges**: Only the information about charges is included. The output file format reads

$$1.\eta_s \quad 2.x \quad 3.y \quad 4.(\tau\epsilon_g)_0 \quad 5.(\tau\epsilon_q)_0 \quad 6.(\tau n_u)_0 \quad 7.(\tau n_d)_0 \quad 8.(\tau n_s)_0$$
 - * **EMoments**: Includes the moments of the energy distribution. Say we define the moments of this distribution as

$$\begin{aligned}
 E_{I,i,n} &= \int d^2\mathbf{r}(\tau\epsilon_i)_0 |\mathbf{r}|^n \\
 E_{C,i,n} &= \int d^2\mathbf{r}(\tau\epsilon_i)_0 |\mathbf{r}|^n \cos(n\varphi) \\
 E_{S,i,n} &= \int d^2\mathbf{r}(\tau\epsilon_i)_0 |\mathbf{r}|^n \sin(n\varphi)
 \end{aligned} \tag{7}$$

Additionally the quantity

$$E_{2/3} = \int d^2\mathbf{r} [K_g(\tau\epsilon_g)_0 + (\tau\epsilon_q)_0]^{2/3} \tag{8}$$

here $i = g, q$ and $n = 0, 1, 2, \dots$. The default maximum moment is $n = 4$. The format of the file is then given by

$$1.\eta_s \quad 2.E_{2/3} \quad 3.E_{4/3} \quad 4.E_{I,g,0} \quad 5.E_{I,q,0} \quad 6.E_{C,g,0} \quad 7.E_{C,q,0} \quad 8.E_{S,g,0} \quad 9.E_{S,q,0} \quad 10.E_{I,g,1} \quad 11.E_{I,q,1} \quad \dots$$

- * **NMoments**: Includes moments of the charges distributions. Now we define the moments of these distributions as

$$\begin{aligned}
 N_{I,f,n} &= \int d^2\mathbf{r}(\tau n_f)_0 |\mathbf{r}|^n \\
 N_{C,f,n} &= \int d^2\mathbf{r}(\tau n_f)_0 |\mathbf{r}|^n \cos(n\varphi) \\
 N_{S,f,n} &= \int d^2\mathbf{r}(\tau n_f)_0 |\mathbf{r}|^n \sin(n\varphi)
 \end{aligned} \tag{9}$$

for each flavour $f = u, d$ ¹. The file for each value of f is printed out using the following format

$$1.\eta_s \quad 2.N_{I,f,0} \quad 3.N_{C,f,0} \quad 4.N_{S,f,0} \quad 5.N_{I,f,1} \quad 6.N_{C,f,1} \quad 7.N_{S,f,1} \dots$$

These options have to be included in a list form, e.g. ["EMoments", "Charges"].

- **PrintAvg**: If set to **True** it prints out the profile of the average event of the run. The MCDIPPER keeps an internal profile which is updated (the average is updated) at each event. The default is **PrintAvg=False**.
- **BoostInvariant**: If set to **True** outputs everything for midrapidity. Default is set to **BoostInvariant=False**.

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- [1] O. Garcia-Montero, H. Elfner, and S. Schlichting, Phys. Rev. C **109**, 044916 (2024), arXiv:2308.11713 [hep-ph].
 - [2] D. d’Enterria and C. Loizides, Ann. Rev. Nucl. Part. Sci. **71**, 315 (2021), arXiv:2011.14909 [hep-ph].
 - [3] O. Garcia-Montero, S. Schlichting, and J. Zhu, Phys. Rev. D **111**, 076029 (2025), arXiv:2501.14872 [nucl-th].
 - [4] A. H. Rezaeian, M. Siddikov, M. Van de Klundert, and R. Venugopalan, Phys. Rev. D **87**, 034002 (2013), arXiv:1212.2974 [hep-ph].

¹ We have included only light quarks since strangeness density is trivially vanishing in our framework. We are working on the implementation of fluctuations, which will eventually show hotspots on the strangeness density, and therefore moments.